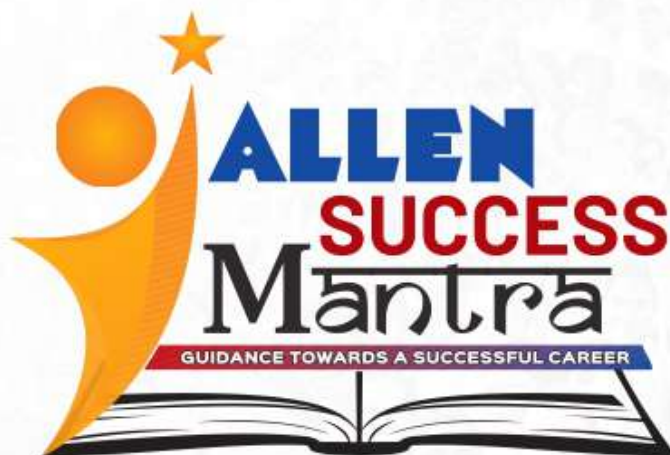


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PARABOLA

1. CONIC SECTIONS :

A conic section, or conic is the locus of a point which moves in a plane so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.

- (a) The fixed point is called the FOCUS.
- (b) The fixed straight line is called the DIRECTRIX.
- (c) The constant ratio is called the ECCENTRICITY denoted by e .
- (d) The line passing through the focus & perpendicular to the directrix is called the AXIS.
- (e) A point of intersection of a conic with its axis is called a VERTEX.

2. GENERAL EQUATION OF A CONIC : FOCAL DIRECTRIX PROPERTY :

The general equation of a conic with focus (p, q) & directrix $lx + my + n = 0$ is :

$$\begin{aligned}
 (l^2 + m^2) [(x - p)^2 + (y - q)^2] &= e^2 (lx + my + n)^2 \\
 &\equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0
 \end{aligned}$$

3. DISTINGUISHING BETWEEN THE CONIC :

The nature of the conic section depends upon the position of the focus S w.r.t. the directrix & also upon the value of the eccentricity e . Two different cases arise.

Case (i) When the focus lies on the directrix :

In this case $D \equiv abc + 2 fgh - af^2 - bg^2 - ch^2 = 0$ & the general equation of a conic represents a pair of straight lines and if :

$e > 1$, $h^2 > ab$ the lines will be real & distinct intersecting at S .

$e = 1$, $h^2 = ab$ the lines will be coincident.

$e < 1$, $h^2 < ab$ the lines will be imaginary.

Case (ii) When the focus does not lie on the directrix :**The conic represents :**

a parabola	an ellipse	a hyperbola	a rectangular hyperbola
$e = 1; D \neq 0$ $h^2 = ab$	$0 < e < 1; D \neq 0$ $h^2 < ab$	$D \neq 0; e > 1$ $h^2 > ab$	$e > 1; D \neq 0$ $h^2 > ab; a + b = 0$

4. PARABOLA :

A parabola is the locus of a point which moves in a plane, such that its distance from a fixed point (focus) is equal to its perpendicular distance from a fixed straight line (directrix).

Standard equation of a parabola is $y^2 = 4ax$. For this parabola :

- (i) Vertex is (0, 0) (ii) Focus is (a, 0)
 (iii) Axis is $y = 0$ (iv) Directrix is $x + a = 0$

(a) Focal distance :

The distance of a point on the parabola from the focus is called the FOCAL DISTANCE OF THE POINT.

(b) Focal chord :

A chord of the parabola, which passes through the focus is called a FOCAL CHORD.

(c) Double ordinate :

A chord of the parabola perpendicular to the axis of the symmetry is called a DOUBLE ORDINATE with respect to axis as diameter.

(d) Latus rectum :

A focal chord perpendicular to the axis of parabola is called the LATUS RECTUM. For $y^2 = 4ax$.

- (i) Length of the latus rectum = $4a$.
 (ii) Length of the semi latus rectum = $2a$.
 (iii) Ends of the latus rectum are $L(a, 2a)$ & $L'(a, -2a)$

Note that :

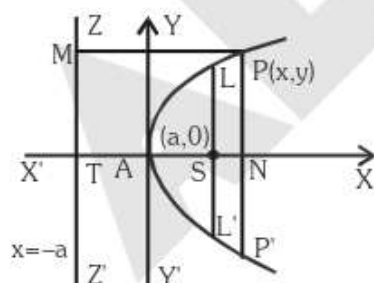
- (i) Perpendicular distance from focus on directrix = half the latus rectum.
- (ii) Vertex is middle point of the focus & the point of intersection of directrix & axis.
- (iii) Two parabolas are said to be equal if they have latus rectum of same length.

5. PARAMETRIC REPRESENTATION :

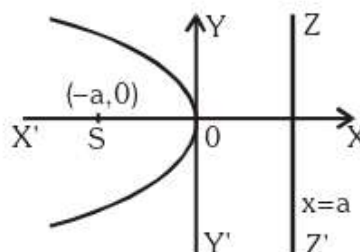
The simplest & the best form of representing the co-ordinates of a point on the parabola $y^2 = 4ax$ is $(at^2, 2at)$. The equation $x = at^2$ & $y = 2at$ together represents the parabola $y^2 = 4ax$, t being the parameter.

6. TYPE OF PARABOLA :

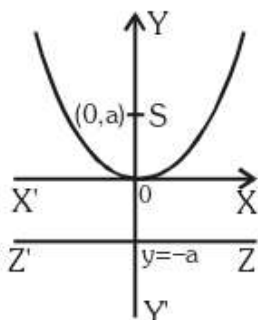
Four standard forms of the parabola are $y^2 = 4ax$; $y^2 = -4ax$; $x^2 = 4ay$; $x^2 = -4ay$



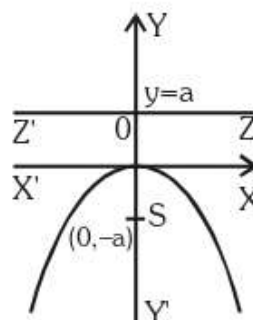
$y^2 = 4ax$



$y^2 = -4ax$



$x^2 = 4ay$



$x^2 = -4ay$

Parabola	Vertex	Focus	Axis	Directrix	Length of Latus rectum	Ends of Latus rectum	Parametric equation	Focal length
$y^2 = 4ax$	(0,0)	(a,0)	$y=0$	$x=-a$	4a	(a, $\pm 2a$)	($at^2, 2at$)	$x+a$
$y^2 = -4ax$	(0,0)	(-a,0)	$y=0$	$x=a$	4a	(-a, $\pm 2a$)	($-at^2, 2at$)	$x-a$
$x^2 = +4ay$	(0,0)	(0,a)	$x=0$	$y=-a$	4a	($\pm 2a, a$)	($2at, at^2$)	$y+a$
$x^2 = -4ay$	(0,0)	(0,-a)	$x=0$	$y=a$	4a	($\pm 2a, -a$)	($2at, -at^2$)	$y-a$
$(y-k)^2 = 4a(x-h)$	(h,k)	(h+a,k)	$y=k$	$x+a-h=0$	4a	(h+a, $k\pm 2a$)	($h+at^2, k+2at$)	$x-h+a$
$(x-p)^2 = 4b(y-q)$	(p,q)	(p, b+q)	$x=p$	$y+b-q=0$	4b	($p\pm 2a, q+a$)	($p+2at, q+at^2$)	$y-q+b$

7. POSITION OF A POINT RELATIVE TO A PARABOLA :

The point (x_1, y_1) lies outside, on or inside the parabola $y^2 = 4ax$ according as the expression $y_1^2 - 4ax_1$ is positive, zero or negative.

8. CHORD JOINING TWO POINTS :

The equation of a chord of the parabola $y^2 = 4ax$ joining its two points $P(t_1)$ and $Q(t_2)$ is $y(t_1 + t_2) = 2x + 2at_1t_2$

Note :

(i) If PQ is focal chord then $t_1t_2 = -1$.

(ii) Extremities of focal chord can be taken as $(at^2, 2at)$ & $\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$

(iii) If $t_1t_2 = k$ then chord always passes a fixed point $(-ka, 0)$.

9. LINE & A PARABOLA :

(a) The line $y = mx + c$ meets the parabola $y^2 = 4ax$ in two points real, coincident or imaginary according as $a \geq < cm$

$\Rightarrow$ condition of tangency is, $c = \frac{a}{m}$.

Note : Line $y = mx + c$ will be tangent to parabola $x^2 = 4ay$ if $c = -am^2$.

(b) Length of the chord intercepted by the parabola $y^2 = 4ax$ on

the line $y = mx + c$ is : $\left(\frac{4}{m^2}\right)\sqrt{a(1+m^2)(a-mc)}$.

Note : length of the focal chord making an angle α with the x-axis is $4a \operatorname{cosec}^2 \alpha$.

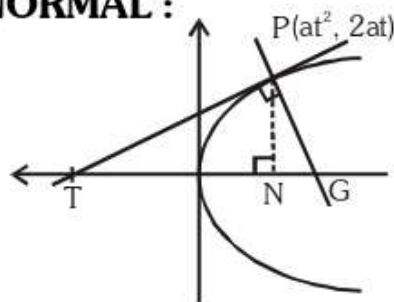
10. LENGTH OF SUBTANGENT & SUBNORMAL :

PT and PG are the tangent and normal respectively at the point P to the parabola $y^2 = 4ax$. Then

TN = length of subtangent = twice the abscissa of the point P

(Subtangent is always bisected by the vertex)

NG = length of subnormal which is constant for all points on the parabola & equal to its semilatus rectum ($2a$).



11. TANGENT TO THE PARABOLA $y^2 = 4ax$:

(a) Point form :

Equation of tangent to the given parabola at its point (x_1, y_1) is $yy_1 = 2a(x + x_1)$

(b) Slope form :

Equation of tangent to the given parabola whose slope is 'm', is

$$y = mx + \frac{a}{m}, (m \neq 0)$$

Point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m} \right)$

(c) Parametric form :

Equation of tangent to the given parabola at its point $P(t)$, is $ty = x + at^2$

Note : Point of intersection of the tangents at the point t_1 & t_2 is $[at_1 t_2, a(t_1 + t_2)]$. (i.e. G.M. and A.M. of abscissae and ordinates of the points)

12. NORMAL TO THE PARABOLA $y^2 = 4ax$:

(a) Point form :

Equation of normal to the given parabola at its point (x_1, y_1) is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1)$$

(b) Slope form :

Equation of normal to the given parabola whose slope is 'm', is $y = mx - 2am - am^3$ foot of the normal is $(am^2, -2am)$

(c) Parametric form :

Equation of normal to the given parabola at its point $P(t)$, is
 $y + tx = 2at + at^3$

Note :

- (i) Point of intersection of normals at t_1 & t_2 is
 $(a(t_1^2 + t_2^2 + t_1t_2 + 2), -at_1t_2(t_1 + t_2))$.
- (ii) If the normal to the parabola $y^2 = 4ax$ at the point t_1 , meets the parabola again at the point t_2 , then

$$t_2 = -\left(t_1 + \frac{2}{t_1}\right).$$
- (iii) If the normals to the parabola $y^2 = 4ax$ at the points t_1 & t_2 intersect again on the parabola at the point ' t_3 ' then
 $t_1t_2 = 2$; $t_3 = -(t_1 + t_2)$ and the line joining t_1 & t_2 passes through a fixed point $(-2a, 0)$.

13. PAIR OF TANGENTS :

The equation of the pair of tangents which can be drawn from any point $P(x_1, y_1)$ outside the parabola to the parabola $y^2 = 4ax$ is given by : $SS_1 = T^2$, where :

$$S \equiv y^2 - 4ax ; \quad S_1 \equiv y_1^2 - 4ax_1 ; \quad T \equiv yy_1 - 2a(x + x_1).$$

14. CHORD OF CONTACT :

Equation of the chord of contact of tangents drawn from a point $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$

Remember that the area of the triangle formed by the tangents from

the point (x_1, y_1) & the chord of contact is $\frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$. Also note

that the chord of contact exists only if the point P is not inside.

15. CHORD WITH A GIVEN MIDDLE POINT :

Equation of the chord of the parabola $y^2 = 4ax$ whose middle point

is (x_1, y_1) is $y - y_1 = \frac{2a}{y_1}(x - x_1)$.

This reduced to $T = S_1$

where $T \equiv yy_1 - 2a(x + x_1)$ & $S_1 \equiv y_1^2 - 4ax_1$.

16. DIAMETER :

The locus of the middle points of a system of parallel chords of a Parabola is called a DIAMETER. Equation to the diameter of a parabola is $y = 2a/m$, where m = slope of parallel chords.

17. CONORMAL POINTS :

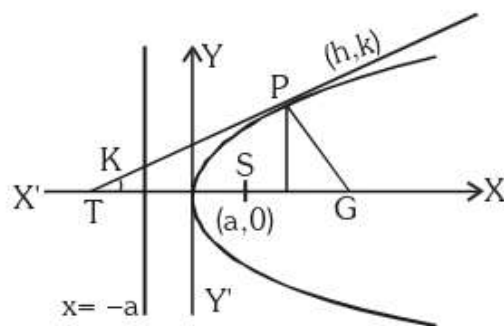
Foot of the normals of three concurrent normals are called conormals point.

- Algebraic sum of the slopes of three concurrent normals of parabola $y^2 = 4ax$ is zero.
- Sum of ordinates of the three conormal points on the parabola $y^2 = 4ax$ is zero.
- Centroid of the triangle formed by three co-normal points lies on the axis of parabola.
- If $27ak^2 < 4(h - 2a)^3$ satisfied then three real and distinct normal are drawn from point (h, k) on parabola $y^2 = 4ax$.
- If three normals are drawn from point $(h, 0)$ on parabola $y^2 = 4ax$, then $h > 2a$ and one of the normal is axis of the parabola and other two are equally inclined to the axis of the parabola.

18. IMPORTANT HIGHLIGHTS :

- (a) If the tangent & normal at any point 'P' of the parabola intersect the axis at T & G then $ST = SG = SP$ where 'S' is the focus. In other words the tangent and the normal at

a point P on the parabola are the bisectors of the angle between



the focal radius SP & the perpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of the parabola after reflection.

- (b) The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the **focus**.
- (c) The tangents at the extremities of a focal chord intersect at right angles on the **directrix**, and a circle on any focal chord as diameter touches the directrix. Also a circle on any focal radii of a point P ($at^2, 2at$) as diameter touches the tangent at the vertex and intercepts a chord of length $a\sqrt{1+t^2}$ on a normal at the point P.
- (d) Any tangent to a parabola & the perpendicular on it from the focus meet on the tangent at the vertex.
- (e) Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord

$$\text{i.e. } 2a = \frac{2bc}{b+c} \quad \text{or} \quad \frac{1}{b} + \frac{1}{c} = \frac{1}{a}.$$

- (f) Image of the focus lies on directrix with respect to any tangent of parabola $y^2 = 4ax$.